# 01

# NEARBY

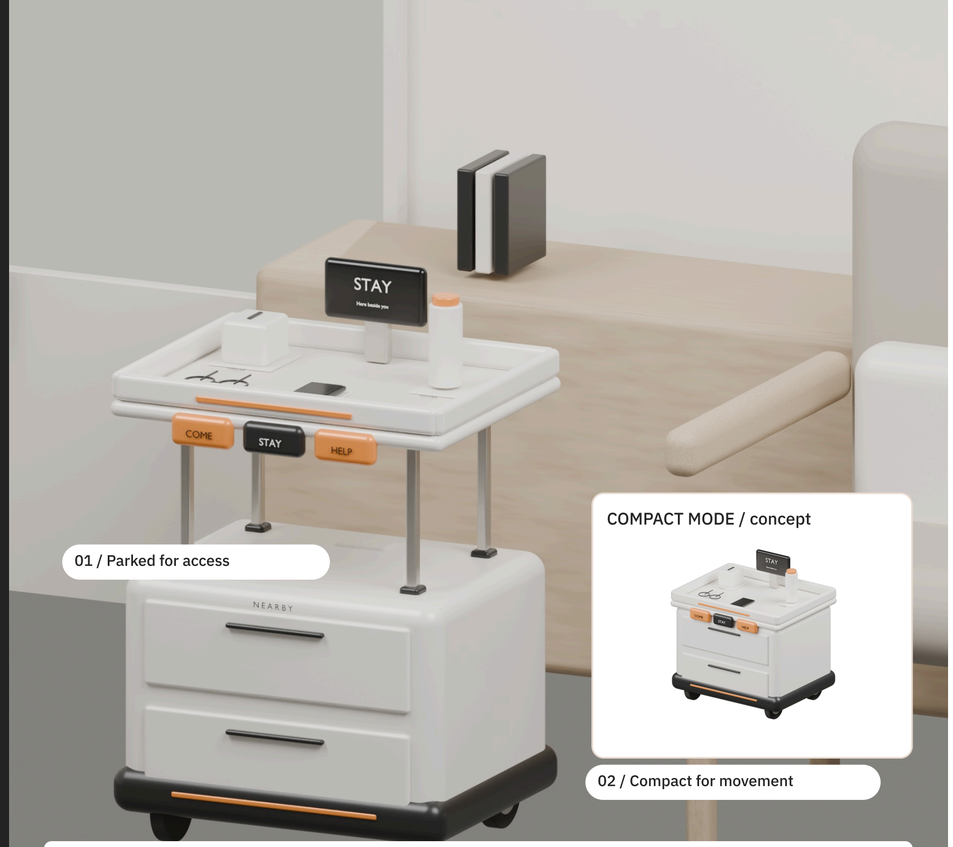
## Familiar things.
## Within reach.

A home companion cabinet for people living with dementia and the people who support them.

Fixed places for essentials.
Requested movement.
A clear way to ask for help.

**PRODUCT + INTERACTION + SERVICE**

New home-care concept.
Informed by an early lift-box demo.




Blender concept scene • compact and parked states • new care use to validate

# Background

### What need does NEARBY address?

Misplaced everyday objects can become a source of frustration for a person with dementia. Familiar, accessible places can help support daily routines. [D1]

**NEARBY explores whether those familiar places can move nearby without making the home harder to understand.**

### Home essentials / a proposed fixed order

Glasses · Phone · Water · Tissues

### Design principles

01 **Same slot**
An item returns to the same visible place.

02 **Familiar stop**
The cabinet parks at agreed home locations.

03 **Clear state**
The person can tell when it is stationary.

### Current difficulties / research-led hypotheses

**Searching**
Objects are no longer where expected.

**Reaching**
Essentials may be in another room.

**Requesting**
Complex controls can be hard to interpret.

### Opportunity / mobility with consistency

**Keep the contents predictable**
Adapt the cabinet's location while preserving the arrangement of the person's familiar belongings.

COME → STAY → HELP

Requested movement · Stopped for access · Contact a caregiver

### A shared support system

**Person** → **Cabinet** → **Caregiver**

Preferences + consent / familiar essentials / setup + recovery


[D1] Alzheimer's Society / Losing and misplacing things
[D2] NHS / Dementia-friendly home environment
[D3] NIA / Alzheimer's caregiving: home safety

# Field Research

Secondary research + proposed home inquiry. The following needs and journey are hypotheses, not completed fieldwork.

## Pain-point classification / what to investigate

### Human–Object / Finding familiar things




**Stable item positions may support recognition. [D1]**

Study question: does a moving cabinet preserve that familiarity, or make searching harder?

### Human–Space / Moving through the home




**Lighting, contrast and understandable labels matter in the home. [D2]**

Study question: where can the cabinet stop without narrowing a walking route?

### Human–Caregiver / Returning & replenishing



**Agree slots, replenish items and handle exceptions.**

Study question: does upkeep reduce or add caregiver work?

Illustrations: project Blender scene + editable diagrams.

## User Research & Insights

### Stakeholders / provisional profiles

**Person with dementia**

Can express preferences and try simple cues with agreed support.

**Family caregiver**

Maintains essentials and checks movement or help requests.

**Care professional**

Advises suitable use, accessibility and study boundaries.

### Need reframing / design hypotheses

**Recognize**

Keep an understandable, consistent layout; do not depend on remembering an app.

**Reach**

Bring familiar essentials closer after a clear request, then remain stationary.

**Reassure**

Explain stopped and help-delivery states through simple words and cues.

### Daily Journey / a scenario to validate, not an observed timeline

| Morning | Move room | Arrive | Use items | Need support | Caregiver check |
|---|---|---|---|---|---|
| At a familiar chair | Leave the chair | Reach a stopping point | Take a daily essential | Press HELP | Replenish + inspect |
| Locate glasses | Request the cabinet | Wait until parked | Return it after use | Know if it was sent | Keep the setup familiar |
| Same visible slot | Known target + route | Clear STAY cue | Unchanged tray order | Delivery / failure cue | Adjust with the person |

**Design opportunity: mobility only has value if familiarity, walking space and caregiver workload remain acceptable.**

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# Design Concept

A familiar arrangement, closer by request

**Keep the item positions predictable. Change proximity only when the person wants the cabinet nearby.**

**FIND**
Recognize the familiar slots.

**COME**
Request movement with a simple remote.

**STAY**
Stop before reaching for items.

**HELP**
Ask a trusted caregiver for support.

These controls and care routines belong to the new concept. Their comprehension and acceptance require co-design.

NEW CONCEPT / demo-informed form

# Form Development

Lift-box principle → a domestic cabinet



Front / raised | Side / raised | Perspective

Digital form studies generated from the editable model; newly authored exploration, not historical hand sketches.

CMF / quiet surfaces + clear action accents

Warm porcelain | Satin aluminum | Soft graphite | Warm orange

Rounded cabinet and tray edges; enclosed low wheel base; a small status display extending the demo's mounted-phone idea.

# Interaction Logic

State logic / requested movement, explicit recovery

**READY**
Stationary; request enabled

→

**MOVING**
Target + path confirmed

→

**PARKED**
Stationary for item access

**STOPPED / STAY, obstruction or unknown target**
Explain the stop and wait for an explicit new request. Following, identity checks and path sensing still require engineering validation.

Test the value of mobility against simpler options

**Static organizer**
Predictable positions. May require returning to one place.

**Manual cart**
Repositionable essentials. Needs someone to move it.

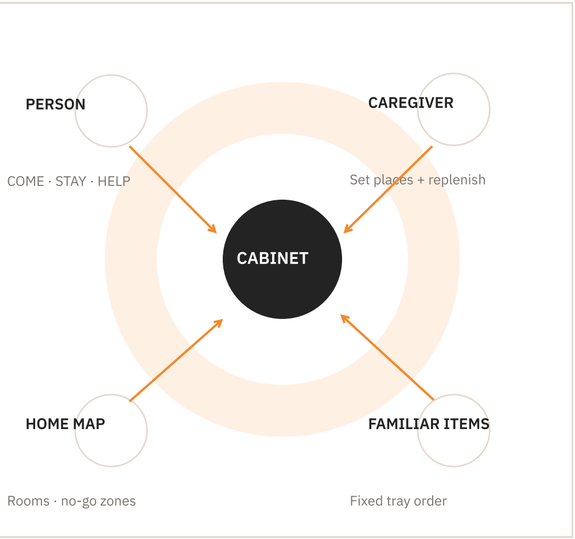
**Following cabinet**
Closer items by request. Adds motion, upkeep and acceptance questions.

**Continue the mobile concept only if it supports everyday routines better than the simpler alternatives.**

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# System Design

System map / locally anchored routines




**LOCAL / proposed map & route rules**

Physical STOP stays available; the caregiver can revise stops.

01 Family or caregiver labels familiar stopping places.
02 Cabinet follows only a confirmed, clear route.

# Touchpoint Definition

Cabinet + remote / cooperative control




Lift tray · drawers · COME / STAY / HELP

**Personal touchpoint**

Three physical controls

Large, labeled and distinct.

COME | STAY | HELP

**Caregiver app**

Review map, stops, items and requests.

Status at a glance

Proposed sequence / approval before motion

**01 / MAP**
Caregiver guides first pass

**02 / LABEL**
Name agreed parking spots

**03 / REVIEW**
Mark no-go and clear paths

**04 / MOVE**
Requested route; stop if blocked

**SPECIFICATION, NOT A TESTED CAPABILITY**
Mapping, localization, obstacle detection and path safety still require implementation and home testing.

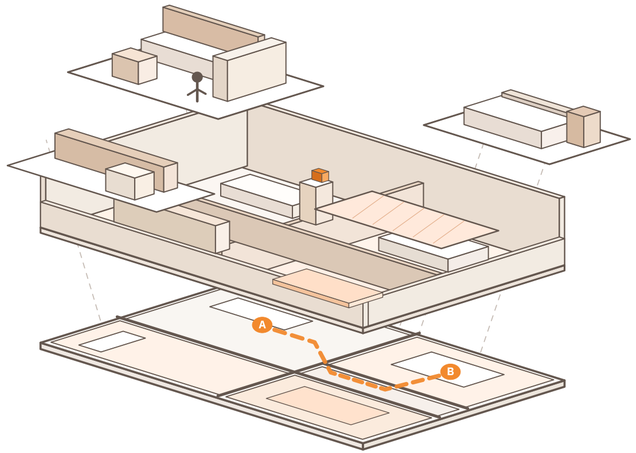
# Touchpoints + Environment

Home map / simulated, proposed feature

**EXPLODED HOME / CONCEPT**

01 / ROOM COMPONENTS

02 / OPEN-WALL MODEL




03 / ROUTE PLAN

A + B / agreed stops — Proposed route — Bathroom / no-go

Axonometric home study. Navigation map and obstacle handling are design proposals, not sensor output.

[M1] Nav2 / Mapping & Localization
[M2] Nav2 / Keepout Filter

# User Flow & Interface

Direct actions for the person; caregiver-reviewed home map, familiar stops and recovery.

## User flow / proposed interaction logic

**Choose together**
Agree familiar essentials, cues and preferred support.

→

**Map home together**
Label rooms and safe routes with the caregiver.

→

**Review familiar stops**
Mark restricted rooms and agree where to park.

**Request COME**
The remote acknowledges the request. Confirm target + path.

→

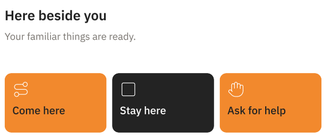
**Move / STAY available**
Unknown target, obstruction or STAY → stop; no automatic restart.

→

**Park → use items**
Show a clear stationary state before reaching for essentials.

**HELP / a separate contact flow**
Request → sending → delivered or failed → caregiver acknowledges. Failed delivery needs a clear fallback agreed during setup.

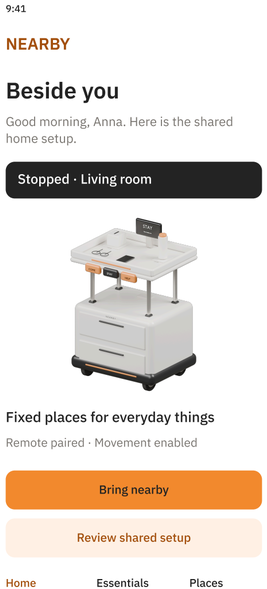
## Person-facing remote / direct actions



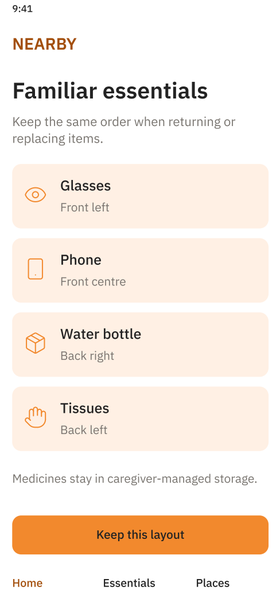

Paired remote + cabinet controls
COME / STAY / HELP
Large words, icons and distinct states;
cue preferences require co-design.

## Caregiver app / editable high-fidelity screens

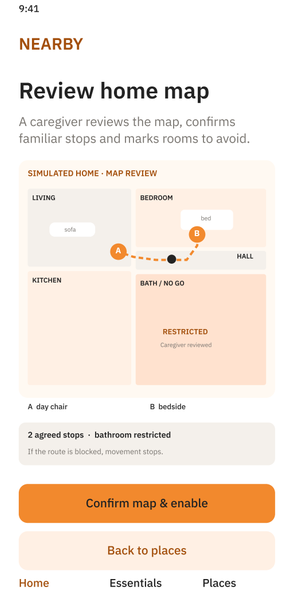
### Home



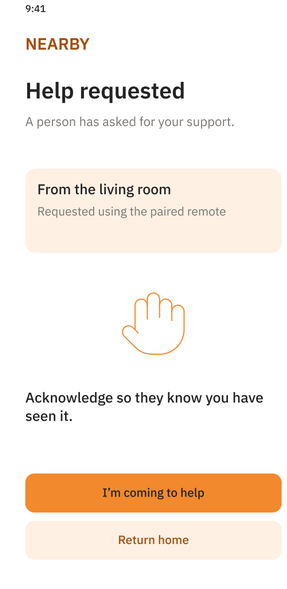

### Essentials




### Home map




### Help request



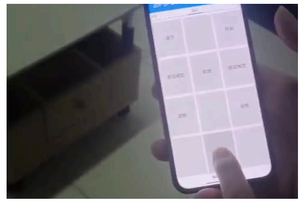


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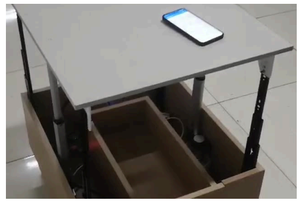
# Prototype, Service & Validation

An early technical attempt anchors the new project; home-care suitability and motion performance remain to be tested.

## 01 / Original demo attempt



Remote commands



Raised surface



**Storage box + rising platform + mounted phone**

The supplied video documents a technical exploration. It informed form and motion, but did not test dementia-care use or verify autonomous following.

## 02 / A new household service routine

| | **Agree**<br>Preferences + consent | **Set up**<br>Slots + familiar stops | **Use**<br>Request → park → reach | **Maintain**<br>Refill + inspect |
|---|---|---|---|---|
| **Person** | Choose familiar items | Try cues and controls | Use items while stopped | Return to the same slots |
| **Caregiver** | Agree appropriate support | Pair remote and target | Check exceptions | Refill; charge; inspect |
| **Cabinet** | No default movement | Store agreed settings | Show clear status | Report upkeep needs |
| **Service** | Assess suitability | Provide home guidance | Help troubleshoot | Repair; support an exit |

## Service model / a hypothesis to explore

Explore setup-supported rental or purchase and repairable parts. Pricing, acceptance and caregiver workload remain untested.

## 03 / Validation plan, not results

### Co-design first

Discuss meaningful objects, preferred cues and acceptable movement with the person, caregiver and care professionals.

### Compare simple alternatives

Use the same item tasks to compare static storage, a manual cart and a requested-following cabinet.

### Observe everyday outcomes

Finding and returning items; interpreting motion and help status; blocked routes; maintenance; acceptance or refusal.

### Simplify when needed

If mobility adds confusion, near misses or unwanted work, reduce automation or return to static storage.

Demo stills: supplied video. All care scenarios and new product features are design proposals.

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